

EngiLab Frame.2D 2022 Analysis Report

Model: "Model1.fr2d"

Report created: Tuesday, 06 June 2023, 16:50:07

Node Displacements

Node ID	X-Displ. [mm]	Y-Displ. [mm]	Z-Rot. [deg]
1 (A)	0.00	0.00	0.00
2 (B)	37.16	-0.04	-1.18
3 (C)	37.16	-43.62	-1.24
4 (D)	37.16	-44.59	-1.24

Support Reactions

Node ID	FX-Force [kN]	FY-Force [kN]	MZ-Moment [kN·m]
1 (A)	0.00	7.30	14.97

Background color: Free DOF Constrained DOF Spring DOF

Element End Forces

Element ID	Axial F. [kN]	Shear F. [kN]	Bend. M. [kN·m]
1 (A) Start	-7.30	0.00	-14.97
1 (A) End	-7.30	0.00	-14.97
2 (B) Start	0.00	7.30	-14.97
2 (B) End	0.00	7.30	0.00
3 (C) Start	0.00	0.00	0.00
3 (C) End	0.00	0.00	0.00

Background color: Normal DOF Hinge DOF

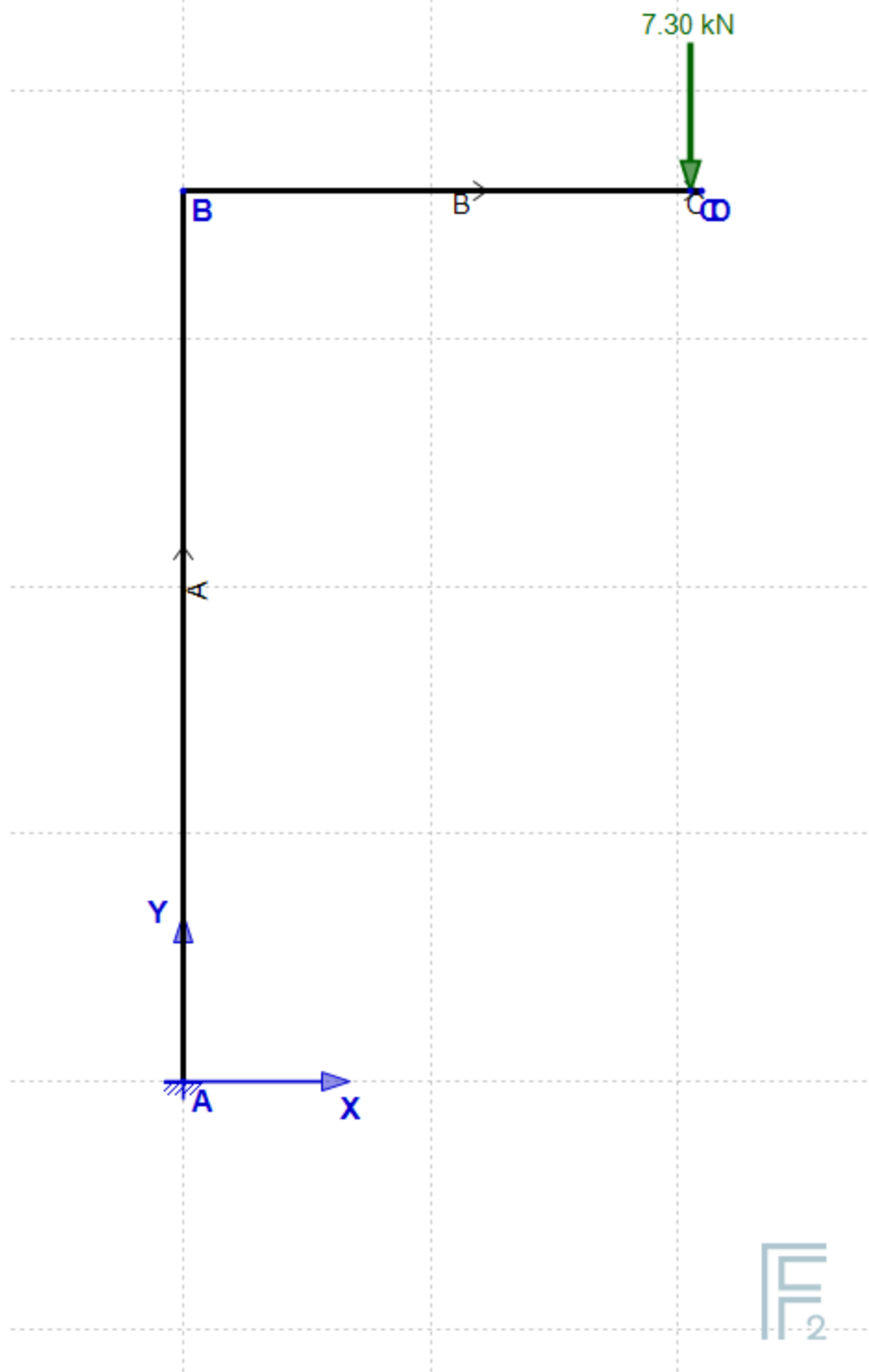
Sign convention: According to the diagrams' sign convention

Axial Force: Positive when member is in tension

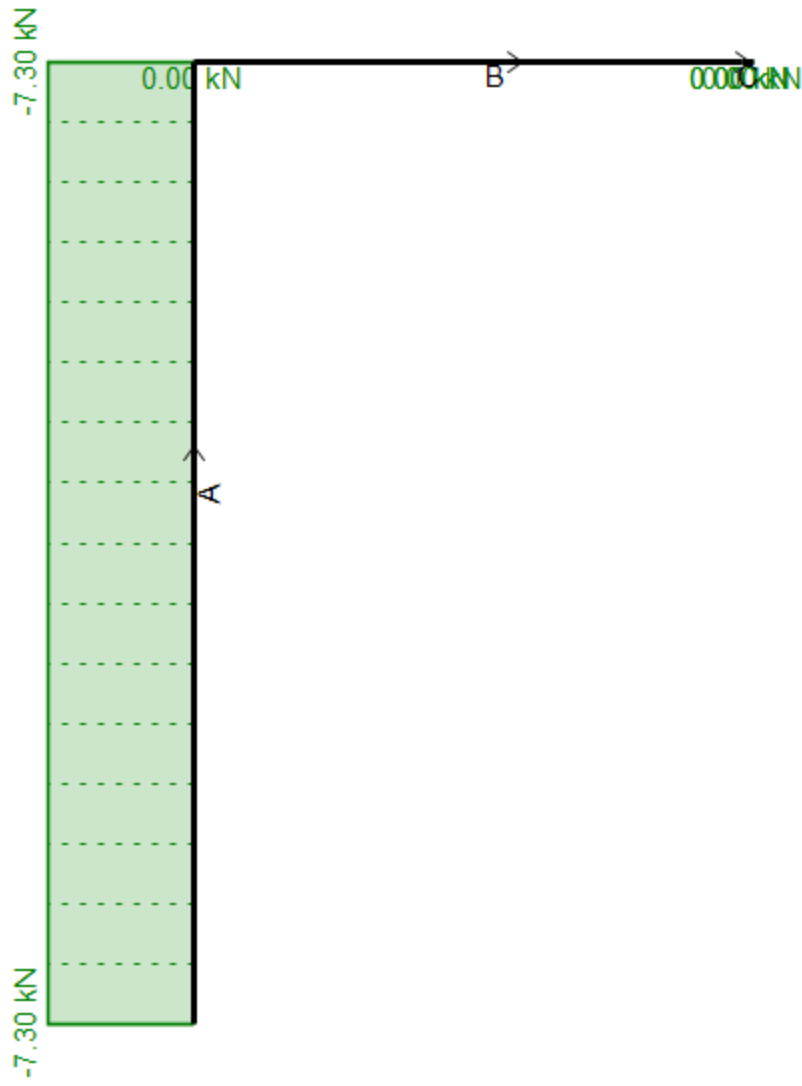
Shear Force: Positive when it rotates the member clockwise

Bending Moment: Positive when the "bottom" fiber is in tension ("top" fiber in compression)

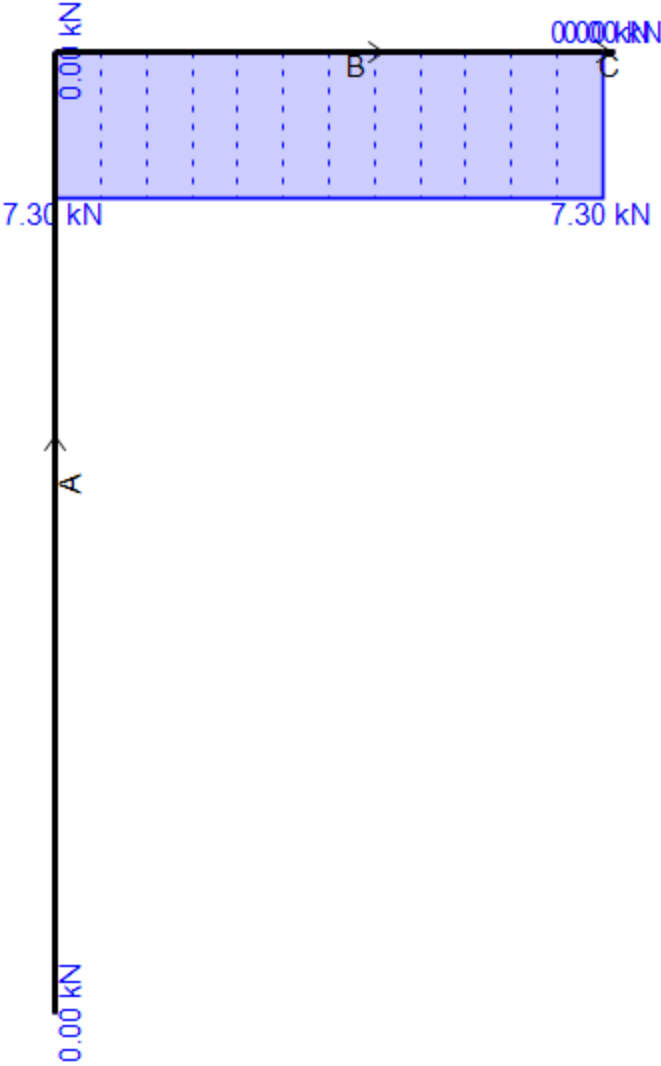
Model



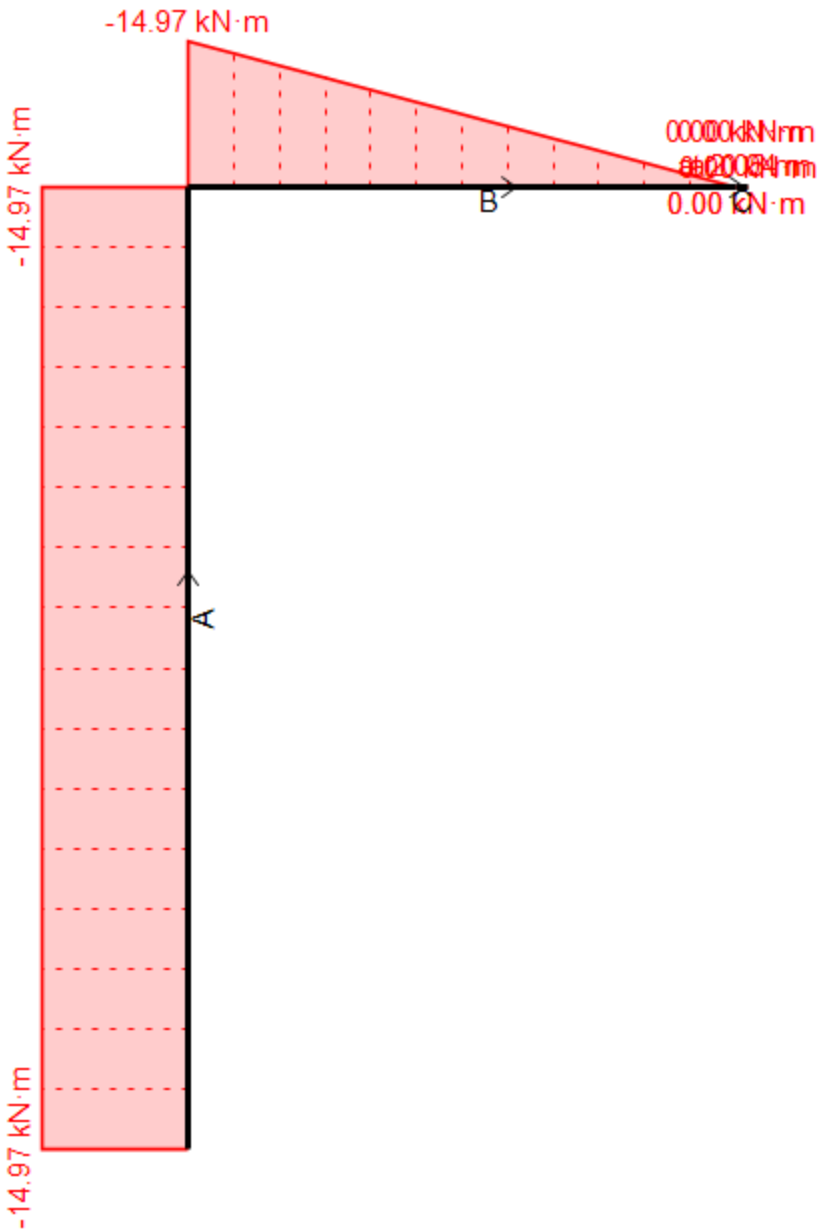
Axial Force Diagram [N]



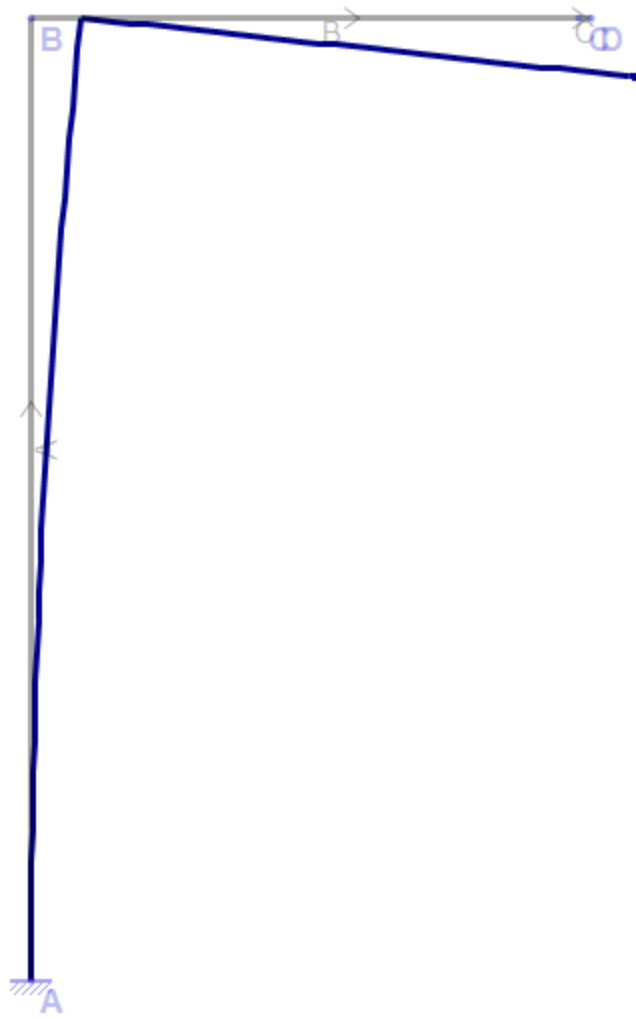
Shear Force Diagram [V]



Bending Moment Diagram [M]



Deformation
(x5.03)



Free Body Diagram

Model Equilibrium

$$\Sigma F_X = 0.00 \text{ kN}$$

$$\Sigma F_Y = 0.00 \text{ kN}$$

$$\Sigma M_Z(0,0) = 0.00 \text{ kN}\cdot\text{m}$$

